Quantum Coding LLM Research Report
Gemma 4 26B-A4B MoE Warmup Readiness Update

quantum-gpt workspace

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Abstract

This report records the latest verified state of the quantum coding LLM R&D loop. As of 2026-04-09, the key update is not that Gemma training has already started, but that the main line has converged from the high-friction Gemma 4 31B-it path to the more executable Gemma 4 26B-A4B-it MoE path. The local eval gate, strict holdout integrity, local model snapshot, and paper-router warmup data are all verified. A fresh experiment in this turn further sharpened the blocker: the Gemma smoke failed at `runtime_compat`, showing that local `transformers4.49.0` does not recognize the `gemma4` architecture, while the checkpoint metadata advertises `5.5.0.dev0`. The next credible step is therefore to upgrade the local Gemma runtime first, then re-run smoke, and only then sync to ai2 and launch the first router warmup.

1 Executive Summary

As of 2026-04-09, the most accurate project state is:

1. The local R&D loop remains healthy, and the main eval gate stays green after recent suite expansion.
2. The main-line target has converged from the higher-friction `Gemma431B-it` route to the more executable `google/gemma-4-26B-A4B-it` MoE route.
3. The local `Gemma426B-A4B-it` snapshot, paper-router warmup data, and controller command sheet are all in place.
4. The real blocker is no longer model sourcing or missing weights. It is the local Gemma runtime and trainer/backend preflight.

In one sentence:

> The project has moved from chasing bigger models and broad narrative claims to executing the first bounded, falsifiable MoE warmup experiment around Gemma 4 26B-A4B.

2 Verified Facts In This Iteration

2.1 Local Eval Gate

Executed in this turn:

```
python3 evals/runner/run_eval.py
```

Result:

- Overall:26/26passed
- quantum:13/13passed
- software:13/13passed

This shows that recent Gemma, Huanxin, dataset, and controller work has not broken the existing capability surface.

2.2 Strict Holdout Integrity

Executed in this turn:

```
python3 scripts/verify_holdout_dataset.py \
  --train-file data/generated/omnicoder-quantum-generalization-holdout-v1/train.jsonl
    \
  --eval-file data/generated/omnicoder-quantum-generalization-holdout-v1/eval.jsonl \
  --manifest data/generated/omnicoder-quantum-generalization-holdout-v1/manifest.json
    \
  --min-eval-count 500 \
  --require-task-disjoint \
  --require-prompt-family-disjoint
```

Result:

- train: 1024
- eval: 504
- example_id overlap: 0
- task_id overlap: 0
- prompt_family overlap: 0
- ok:true

The current main line therefore still stands on a strict unseen-eval framework instead of regressing toward leakier train/eval practice.

2.3 Gemma 4 26B-A4B Local Snapshot

Executed in this turn:

```
python3 training/verify_qwen_snapshot.py \
  models/gemma-4-26B-A4B-it \
  --expected-substring gemma-4-26B-A4B-it \
  --expected-family-substring gemma
```

Result:

- status:ok
- config_model_type:gemma4
- config_architectures:["Gemma4ForConditionalGeneration"]
- Full weight shard set present:
  - model-00001-of-00002.safetensors

  - model-00002-of-00002.safetensors
  - model.safetensors.index.json

- Processor and template artifacts present:
  - processor_config.json
  - chat_template.jinja

The problem is therefore no longer "the model is not fully downloaded." The problem is now "the model is local, but the runtime still has to prove it can actually load and use it."

2.4 Gemma Smoke: Fresh Blocker In This Turn

Executed in this turn:

```
python3 training/huanxin_cpu_smoke.py \
  --model-name models/gemma-4-26B-A4B-it \
  --dataset data/seed/splits-auto-seed/train.jsonl \
  --max-samples 1
```

The command failed, and the blocker is now much more specific:

- stage: `runtime_compat`
- status: `error`
- local `transformers`: `4.49.0`
- checkpoint `transformers_version`: `5.5.0.dev0`
- `AutoConfig` cannot recognize `model_type=gemma4`
- the error also makes it explicit that even after a runtime upgrade, the current text-only `AutoModelForCausalLM` path may still need a processor-aware conditional-generation backend

This matters because it upgrades the older generic statement "Gemma trainer/backend preflight is not green yet" into a more executable problem:

> The first blocker is that the local Gemma runtime is too old. The second blocker is whether the training path still needs a stronger conditional-generation backend after that upgrade.

2.5 Paper-Router Warmup Data

Executed in this turn:

```
wc -l \
  data/generated/quantum-paper-router-warmup-v1/messages/
    quantum_paper_router_warmup_messages_train.jsonl \
  data/generated/quantum-paper-router-warmup-v1/messages/
    quantum_paper_router_warmup_messages_valid.jsonl
```

Result:

- train rows: `42`
- valid rows: `5`

This dataset is not large enough to support a final capability claim, but it is large enough for a very specific first experiment: use paper-domain text for router warmup, establish a domain routing bias in the MoE, and only then decide whether to proceed to expert selection and refinement.

3 Why The Main Line Shifted To Gemma 4 26B-A4B

The main line converged on `google/gemma-4-26B-A4B-it` for engineering reasons rather than narrative reasons:

1. The `31B` route carries too much friction in download, sync, and runtime preparation.
2. The `26B-A4B` MoE structure is a better fit for router warmup, expert selection, and bounded ESFT or LoRA-on-MoE experiments.
3. The current repo already aligns the following assets to this line:
   - local snapshot verification
   - paper-router data
   - controller reports
   - ai2 queue scripts
   - the exact warmup command sheet

The most valuable next outcome is therefore not a longer argument for why 31B might eventually be stronger, but the first real warmup evidence on `26B-A4B`.

4 Current Research Method

The project still follows the same engineering-driven method:

1. Start from executable tasks and verifiable data instead of vague capability slogans.
2. Pass local eval and structural checks before remote training.
3. Compress each blocker into the smallest executable, attributable, and recordable problem.
4. For the Gemma MoE line, prioritize:
   - `routerwarmup`
   - `routingfrequencyinspection`
   - `selectedexpertrefinement`

The point is not simply to move fast. The point is to make each step falsifiable enough that the loop does not fool itself.

5 Most Credible Next Actions

The smallest, cleanest, and most evidence-aligned next steps are:

1. Upgrade the local Gemma runtime so that it can at least recognize `gemma4`.
2. Re-run local smoke and confirm whether the blocker moves from `runtime_compat` to a more specific conditional-generation backend issue.

3. If smoke passes, execute the ai2 sync chain:

```
scripts/push_to_s3.sh scripts training evals research data
scripts/ai2_sync_from_s3.sh
scripts/ai2_sync_model_from_s3.sh gemma-4-26B-A4B-it
```

4. Then launch the paper-router warmup:

```
PYTORCH_NPU_ALLOC_CONF=max_split_size_mb:256 \
TOKENIZERS_PARALLELISM=false \
ASCEND_RT_VISIBLE_DEVICES=0,1,2,3,4,5,6,7 \
torchrun --nproc_per_node=8 training/qwen_sft_peft.py \
  --model-name models/gemma-4-26B-A4B-it \
  --train-file data/generated/quantum-paper-router-warmup-v1/messages/
    quantum_paper_router_warmup_messages_train.jsonl \
  --eval-file data/generated/quantum-paper-router-warmup-v1/messages/
    quantum_paper_router_warmup_messages_valid.jsonl \
  --output-dir outputs/gemma4-26b-a4b-it-quantum-paper-router-warmup \
  --device npu \
  --max-length 1024 \
  --per-device-batch-size 1 \
  --gradient-accumulation-steps 2 \
  --learning-rate 1e-4 \
  --num-epochs 1 \
  --max-steps 20 \
  --eval-steps 1000 \
  --log-steps 5 \
  --train-on-completions-only \
  --target-module-regex '(?:^|\.)(?:router|gate)(?:$|\.)' \
  --trainable-param-regex 'lora_'
```

This sequence is valuable because every step has a clear truth boundary, any failure can be attributed precisely, and a success would produce the first real training evidence for the Gemma MoE line.

6 Related Artifacts

The core artifacts directly attached to the current main line are:

Controller reports:

```
reports/autonomous_rd_cycle_gemma4-26b-a4b-it_2026-04-09.md
reports/autonomous_rd_cycle_gemma4-26b-a4b-it_2026-04-09.json
reports/autonomous_rd_cycle_gemma4-26b-a4b-it_2026-04-09_state.json
```

Controller command sheet:

```
artifacts/autonomous_rd_cycle_gemma4-26b-a4b-it_2026-04-09_commands.txt
```

Paper-router data:

```
data/generated/quantum-paper-router-warmup-v1/messages/
    quantum_paper_router_warmup_messages_train.jsonl
data/generated/quantum-paper-router-warmup-v1/messages/
    quantum_paper_router_warmup_messages_valid.jsonl
```

These are evidence files that support the current narrative. They are not separate maintained versions of the human-facing research report.

7 Final Assessment

The most accurate statement about this iteration is:

> The Gemma 4 26B-A4B line has moved from a conceptual target to a state where most evidence is already in place and the first MoE warmup is waiting on one concrete gating fix. The local eval gate, strict holdout, model snapshot, paper warmup data, and controller command sheet are all ready; the fresh experiment in this turn shows that the first blocker is an outdated local `transformers` runtime, and the second blocker is the eventual conditional-generation backend adaptation.

This LaTeX source is now the single maintained research report. Dated controller outputs remain run artifacts only.